

Vedant Ratn Nema

B. Tech, Computer Science, VIT Vellore
CFA-L1 (In-pursuit)

✉ vedantrnema05@gmail.com

☎ +91-7693053622

📍 Bangalore, Karnataka

🐙 <https://github.com/Vedant-Ratn-Nema>

in <https://www.linkedin.com/in/vedant-ratn-nema>

Professional Experience

Axion Ray – Lead Analytics – Data Science (Mar’ 25 – Present)

- Lead the **Healthcare POD** within the data-ops team, managing multiple enterprise accounts and delivering analytics-driven ROI for clients.
- Spearheaded the **Medtronic engagement**, now signed for **\$600K/2 years**, with projected **\$10M+ annual ROI** through predictive analytics on medical device complaints and recall risk modeling.
- Partnered with **Medline** on a **\$50K/month pilot** projecting **\$6M+ annual ROI** by developing a **text matching model** that aligns Amazon reviews with client complaint data, and creating a **sales-to-complaints threshold framework** to identify 100+ high-priority product issues.
- Designed and delivered **custom demos**:
 - **Vantive** – complaint and MDR (Medical Device Reports) analytics.
 - **Olympus** – competitive insights through MDR benchmarking against competitors.
- Built internal automation tools to scale analytics:
 - **MDR Scraper** using OpenFDA API + Python to extract MAUDE records in seconds (vs. slow manual 500 record downloads), now standard across all healthcare accounts.
 - **OCR-enabled PDF Scraper** to process structured/unstructured complaint data, including scanned images.
- Previously, built:
 - **Recall prediction model** (LLM-based) for Medtronic, surfacing complaint-driven manufacturing improvements.
 - **Customer sentiment clustering** with GPT-4o for El-Palacio-de-Hierro, improving brand positivity across social media.

Genpact – Data Scientist (Aug’22 – Nov’24)

Open Pay Electronic Payment Reconciliation (Domain – BFS)

- Architected a tool for reconciling different payments received by Walmart LATAM using similarity matching. Saving the delay of payment verification i.e., **100-man hours per week**.
- Deployed on AppMan as a Tkinter application as well as a Streamlit application

Tax Forecasting – Walmart Mexico/LATAM (Domain – BFS, CPG)

- Created an ML model to predict in a sustainable way the cash flow required for indirect tax in Mexico to increase by 20% which in turn **increased \$50MUSD in cash availability**.
- Hosted the application using Element ML and implemented schedulers for the process accordingly.

PredictPro - Forecasting Tool for Merck (Domain – Pharmaceuticals)

- Introduced and developed a Forecasting tool in Python using the transition curve and various ML Models. Main algorithms included SARIMAX and Exponential Smoothing.
- Deployed this tool by creating Flask API integrated with ReactJS for user interface. Impact created for the Merck forecasting team was saving them **500 hours a week, roughly 50% man-hours**.

Johnson & Johnson (Domain – Pharmaceuticals)

- Played a crucial role in Web scraping public websites related to Medicines and Medical equipment like FDA and US Pharmaceuticals.
- Trained the **LSTM model** for intent classification on the same dataset and made a **Streamlit dashboard** to incorporate this ML Model on the backend which later gave visual outputs on the dashboard

DRDO, CAIR – Machine Learning Intern (June’20 – Oct’20)

Seasoned under the guidance of Scientists of DRDO, Project Comprised of Two Phases.

- Created Data Set: Various screenshots were taken which were later labelled under 10 distinct classes using the Image Label Master
- Developed YOLOv4 model: The data was used to create a satellite image detection and classification system which is **used in various Drone Technology and Missile systems**

Education

B.Tech. Computer Science and Technology
Vellore Institute of Technology, Vellore
2022

Higher Senior Secondary

Central Academy, Jabalpur, M.P.
2018

Technical Skills

- **Python:** Pandas, Numpy, Matplotlib, Keras, TensorFlow, OpenCV
- **Machine Learning:** Supervised, Unsupervised, Predictive Modelling, Forecasting. Hypothesis Testing, Statistics
- **UI:** Tkinter, Streamlit, Flask, ReactJS
- SQL, Java, C/C++
- MS Office Suite, SharePoint, PowerBI, Tableau, MATLAB, R
- **Linux:** SQLMap, Metasploit, Nmap
- **Version Control Tool:** Git

Certifications

- DRDO (CAIR), ML Intern
- Algorithms on Graphs, UC San Diego
- Business Mathematics, VIT, Vellore
- LSM, VIT, Vellore

Personal Projects

- Object Detection using OpenCV and Machine Learning (2022)
- METASPLOIT-Penetration attack on Windows and Android (2021)

Others

- Co-Founder, Young Reformers NGO, Jabalpur, Madhya Pradesh
- Board of Directors, VIT Spartans, VIT Vellore
- Media and Organizing Partner: iCASIC (international Conference on Automation, Signal Processing, Instrumentation and Control), ISA.
- Manager: Stalls and Sponsorship: Gravitas 2020, VIT University, Vellore